SE

Bio-Prep SE 100/17 and 1000/17 Size Exclusion Chromatography Columns

- Prepacked Biocompatible Cross-Linked Agarose Column
- High Mechanical and Chemical Stability
- Fractionation Ranges 5,000–100,000 Da, 10,000–1,000,000 Da
- High Resolution Separations

Summary

Size exclusion chromatography remains one of the most widely used and accepted chromatographic techniques available. This technique is based on the separation of biomolecules according to their molecular weight, and has been extensively used for analytical, preparative, and process scale separations.

Product Description

The Bio-Prep SE 100/17 and SE 1000/17 glass columns are based on biocompatible, 17 µm spherical, crosslinked agarose beads. The beads are produced from highly purified agarose utilizing a proprietary crosslinking process which allows the production of a mechanically and chemically stable material.

Two fractionation ranges are available, 5,000–100,000 Da using the Bio-Prep SE 100/17 column, and 10,000–1,000,000 Da using the Bio-Prep SE 1000/17 column.

Sample volumes of 0.6–2% of the bed volume and flow-rates from 0.1 to 2 ml/min are recommended for normal operation. Larger sample volumes of up to 5% of the bed volume are possible where lower resolution is acceptable *e.g.* desalting.

Typically, lower flow rates and smaller sample volumes yield superior separations. However, the robust nature of these materials makes superior separations at elevated flow rates possible if process throughput is a concern, or during operations such as sanitization, equilibration, etc.

Figure 1. Kd Bio-Prep SE 100/17 gel.

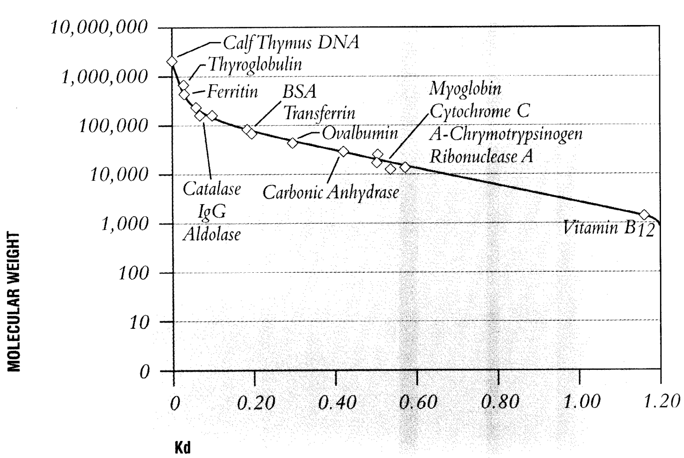


Figure 2. Separation of a gel filtration standard using the SE 100/17 column.

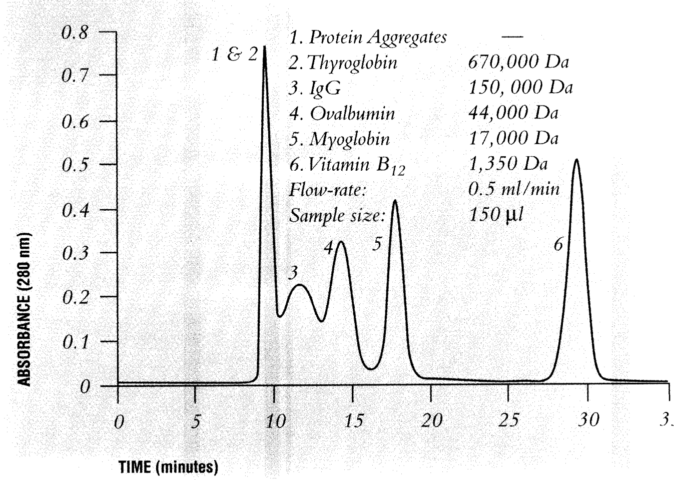


Figure 3. Kd Bio-Prep SE 1000/17 column.

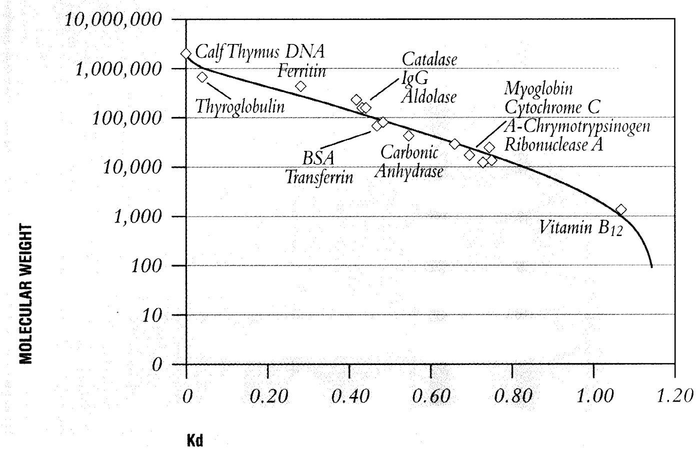


Figure 4. Separation of a gel filtration standard using the Bio-Prep SE 1000/17 column.

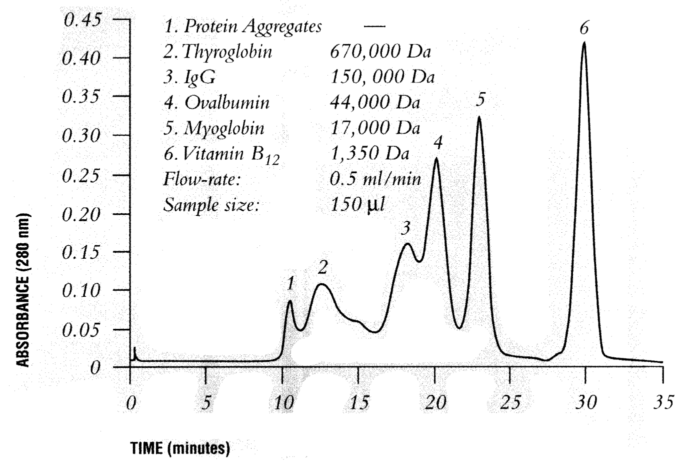
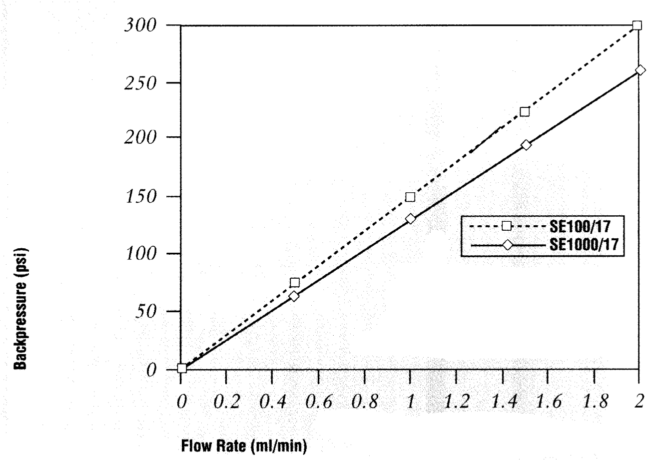


Figure 5. Pressure/Flow of Bio-Prep SE100/17 and SE1000/17, 8 x 300 mm Columns.



Applications

Applications range from the separation of protein aggregates (dimers, trimers, etc.) from purified antibody solutions to the separation of complex mixtures of bio-molecules directly from cell culture feedstreams. Size exclusion chromatography is commonly used as a final polishing step in the preparation of many bio-pharmaceuticals and has numerous applications as a QC tool for assessing product purity, concentration, etc.

Ordering Information

	Bio-Prep SE100/17	Bio-Prep SE1000/17
Catalog Number	**732-1501**	**732-1502**
Column Dimension	8 x 300 mm	8 x 300 mm
Column Housing	Glass	Glass
Bed Matrix	Cross-linked agarose	Cross-linked agarose
Gel Volume	15.0 ml	15.0 ml
Particle Size	15 -20 µm	15 -20 µm
pH Range	1 to 14	1 to 14
Maximum Pressure	580 psi, 40 bar	430 psi, 30 bar
Recommended Flow-rate	0.1 -1.0 ml/min	0.1 -1.0 ml/min
Maximum Flow-rate	2.0 ml/min	2.0 ml/min
Linear Separation Range	10,000 - 100,000 Da	10,000 - 1,000,000 Da
Exclusion Limit	~ 200,000 Da	~ 1,500,000 Da
Cleaning Agent	0.5 m NaOH	0.5 m NaOH

BIO-RAD

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Website *www.bio-rad.com* ***Bio-Rad Laboratories Main Office*** *2000 Alfred Nobel Drive, Hercules, CA 94547, Ph. (510) 741-1000, Fx. (510)741-5800*
Also in: Australia *Ph. 02-9914-2800, Fx. 02-9914-2889* ***Austria*** *Ph. (1)-877 89 01, Fx. (1) 876 56 29* ***Belgium*** *Ph. 09-385 55 11, Fx. 09-385 65 54*
Canada *Ph. (905) 712-2771, Fx. (905) 712-2990* ***China*** *Ph. (86-10) 2046622, Fx. (86-10) 2051876* ***Denmark*** *Ph. 39 17 9947, Fx. 39 27 1698*
Finland *Ph. 90 804 2200, Fx. 90 804 1100* ***France*** *Ph. (1) 43 90 46 90, Fx. (1) 46 71 24 67* ***Germany*** *Ph. 089 31884-0, Fx. 089 31884-100*
Hong Kong *Ph. 7893300, Fx. 7891257* ***India*** *Ph. 91-11-461-0103, Fx. 91-11-461-0765* ***Israel*** *Ph. 03 951 4127, Fx. 03 951 4129*
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Sweden *Ph. 46 (0) 8 627 50 00, Fx. 46 (0) 8 627 54 00* ***Switzerland*** *Ph. 01-809 55 55, Fx. 01-809 55 00*
United Kingdom *Ph. 0800 181134, Fx. 01442 259118*

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